

**COMPUTER NETWORKING
TRANSPORT LAYER PROTOCOL**



KHEN MUHAMMAD CAHYO

244107023002

TI-21

**STATE POLYTECHNIC OF MALANG
INFORMATION TECHNOLOGY DEPARTMENT**

A. Netstat on Linux OS

- Install net-tools

```
debian@debian:~$ sudo apt install net-tools
[sudo] password for debian:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 250 kB of archives.
After this operation, 1,015 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bullseye/main amd64 net-tools amd64 1.60+git20181103.0eebece-1+deb11u1 [250 kB]
Fetched 250 kB in 6s (40.4 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 28252 files and directories currently installed.)
Preparing to unpack .../net-tools_1.60+git20181103.0eebece-1+deb11u1_amd64.deb ...
Unpacking net-tools (1.60+git20181103.0eebece-1+deb11u1) ...
Setting up net-tools (1.60+git20181103.0eebece-1+deb11u1) ...
Processing triggers for man-db (2.9.4-2) ...
debian@debian:~$
```

- Display Open and Listening Port

```
debian@debian:~$ sudo netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*              LISTEN      374/sshd: /usr/sbin
tcp6       0      0 :::22                  :::*                    LISTEN      374/sshd: /usr/sbin
debian@debian:~$ _
```

- Display All Active Connections & Ports

```
debian@debian:~$ netstat -l
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*              LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
raw        0      0 0.0.0.0:icmp            0.0.0.0:*              7
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7

Active UNIX domain sockets (only servers)
Proto RefCnt Flags               Type               State              I-Node  Path
unix   2      [ ACC ] STREAM            LISTENING          12904   /run/user/1000/systemd/private
unix   2      [ ACC ] STREAM            LISTENING          10744   /run/systemd/private
unix   2      [ ACC ] STREAM            LISTENING          10746   /run/systemd/userdb/io.systemd.DynamicUser
unix   2      [ ACC ] STREAM            LISTENING          10747   /run/systemd/io.system.ManagedOOM
unix   2      [ ACC ] STREAM            LISTENING          10755   /run/lvm/lvmpolld.socket
unix   2      [ ACC ] STREAM            LISTENING          10759   /run/systemd/fsck.progress
unix   2      [ ACC ] STREAM            LISTENING          10767   /run/systemd/journal/stdout
unix   2      [ ACC ] SEQPACKET         LISTENING          10769   /run/udev/control
unix   2      [ ACC ] STREAM            LISTENING          10965   /run/systemd/journal/io.systemd.journal
unix   2      [ ACC ] STREAM            LISTENING          12137   /run/dbus/system_bus_socket
debian@debian:~$
```

- **Display Active Ports**

```

20 dropped because of missing route
Icmp:
  3 ICMP messages received
  0 input ICMP message failed
  ICMP input histogram:
    echo replies: 3
  3 ICMP messages sent
  0 ICMP messages failed
  ICMP output histogram:
    echo requests: 3
IcmpMsg:
  InType0: 3
  OutType8: 3
Tcp:
  1 active connection openings
  0 passive connection openings
  0 failed connection attempts
  0 connection resets received
  0 connections established
  108 segments received
  103 segments sent out
  0 segments retransmitted
  0 bad segments received
  0 resets sent
Udp:
  19 packets received
  0 packets to unknown port received
  0 packet receive errors
  30 packets sent
  0 receive buffer errors
  0 send buffer errors
  IgnoredMulti: 110
UdpLite:
TcpExt:
  1 TCP sockets finished time wait in fast timer
  90 packet headers predicted
  3 acknowledgments not containing data payload received
  TCPRecvCoalesce: 40
  TCPOFQueue: 2
  TCPOrigDataSent: 2
  TCPDelivered: 3
IpExt:
  InBcastPkts: 110
  InOctets: 278860
  OutOctets: 7906
  InBcastOctets: 15932
  InNoECTPkts: 384
debian@debian:~$ _

```

- **Display IP Address and Port Number**

```

debian@debian:~$ netstat -g
IPv6/IPv4 Group Memberships
Interface      RefCnt Group
-----
lo              1      all-systems.mcast.net
ens3            1      all-systems.mcast.net
lo              1      ip6-allnodes
lo              1      ff01::1
ens3            1      ff02::1:ff37:8e00
ens3            1      ip6-allnodes
ens3            1      ff01::1
debian@debian:~$ _

```

B. Netstat on Windows OS

- Check Netstat

```
C:\Documents and Settings\XP>netstat

Active Connections

    Proto Local Address          Foreign Address         State
C:\Documents and Settings\XP>_
```

- Show All Active Connections and Ports

```
C:\Documents and Settings\XP>netstat -a

Active Connections

    Proto Local Address          Foreign Address         State
TCP      gns3-winxp:epmap      gns3-winxp:0           LISTENING
TCP      gns3-winxp:microsoft-ds gns3-winxp:0           LISTENING
TCP      gns3-winxp:netbios-ssn gns3-winxp:0           LISTENING
TCP      gns3-winxp:1052       201.181.244.35.bc.googleusercontent.com:https ESTABLISHED
TCP      gns3-winxp:1053       a23-9-199-253.deploy.static.akamaitechnologies.com:https TIME_WAIT
on:htp    TIME_WAIT
TCP      gns3-winxp:1058       sb-in-f104.1e100.net:https ESTABLISHED
TCP      gns3-winxp:1059       si-in-f94.1e100.net:http TIME_WAIT
TCP      gns3-winxp:1062       151.101.131.19:http ESTABLISHED
TCP      gns3-winxp:1067       a23-9-199-253.deploy.static.akamaitechnologies.com:https TIME_WAIT
on:htp    TIME_WAIT
TCP      gns3-winxp:1069       s2-172-242.190.203.detik.com:https TIME_WAIT
TCP      gns3-winxp:1070       s2-172-242.190.203.detik.com:https TIME_WAIT
TCP      gns3-winxp:1076       104.21.37.193:https TIME_WAIT
TCP      gns3-winxp:1077       sa-in-f154.1e100.net:https ESTABLISHED
TCP      gns3-winxp:1078       s2-172-242.190.203.detik.com:https TIME_WAIT
TCP      gns3-winxp:1083       s221-cast-102-221-49-103.detik.com:https TIME_WAIT
AIT
TCP      gns3-winxp:1086       sa-in-f157.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1087       s2-244-242.190.203.detik.com:https TIME_WAIT
TCP      gns3-winxp:1088       sh-in-f97.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1089       sf-in-f95.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1090       si-in-f94.1e100.net:http TIME_WAIT
TCP      gns3-winxp:1091       server-18-67-173-90.cgk51.r.cloudfront.net:http TIME_WAIT
TIME_WAIT
TCP      gns3-winxp:1092       104.21.37.193:https TIME_WAIT
TCP      gns3-winxp:1095       s2-228-242.190.203.detik.com:https TIME_WAIT
TCP      gns3-winxp:1098       104.18.38.233:http TIME_WAIT
TCP      gns3-winxp:1099       sh-in-f94.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1100       sc-in-f139.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1104       sc-in-f139.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1105       si-in-f155.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1105       sb-in-f132.1e100.net:https TIME_WAIT
TCP      gns3-winxp:1111       104.22.53.86:https TIME_WAIT
TCP      gns3-winxp:1112       87.70.96.34.bc.googleusercontent.com:https TIME_WAIT
WAIT
TCP      gns3-winxp:1113       server-18-161-184-8.kul50.r.cloudfront.net:https TIME_WAIT
TIME_WAIT
TCP      gns3-winxp:1116       server-108-138-141-99.cgk51.r.cloudfront.net:https TIME_WAIT
ps TIME_WAIT
TCP      gns3-winxp:1120       sb-in-f156.1e100.net:https ESTABLISHED
TCP      gns3-winxp:1121       sa-in-f132.1e100.net:https ESTABLISHED
TCP      gns3-winxp:1123       sb-in-f156.1e100.net:https ESTABLISHED
TCP      gns3-winxp:1135       123.208.120.34.bc.googleusercontent.com:https ESTABLISHED
start server
```

- Display Network Interface Statistic

```
C:\Documents and Settings\XP>netstat -e

Interface Statistics

    Received Sent
Bytes 5064815 479559
Unicast packets 8588 7748
Non-unicast packets 131 67
Discards 0 0
Errors 0 0
Unknown protocols 0 0
C:\Documents and Settings\XP>
```

- Show Network Details Statistic

```
C:\Documents and Settings\XP>netstat -s

IPv4 Statistics

Packets Received                = 9045
Received Header Errors          = 0
Received Address Errors        = 3
Datagrams Forwarded             = 0
Unknown Protocols Received      = 0
Received Packets Discarded      = 158
Received Packets Delivered      = 8887
Output Requests                 = 8034
Routing Discards                = 0
Discarded Output Packets        = 0
Output Packet No Route          = 0
Reassembly Required             = 0
Reassembly Successful           = 0
Reassembly Failures             = 0
Datagrams Successfully Fragmented = 0
Datagrams Failing Fragmentation = 0
Fragments Created               = 0

ICMPv4 Statistics

                Received      Sent
Messages        4             4
Errors           0             0
Destination Unreachable  0             0
Time Exceeded    0             0
Parameter Problems  0             0
Source Quench    0             0
Redirects        0             0
Echoes           0             4
Echo Replies     4             0
Timestamps       0             0
Timestamp Replies 0             0
Address Masks    0             0
Address Mask Replies 0             0

TCP Statistics for IPv4

Active Opens                = 103
Passive Opens               = 1
Failed Connection Attempts  = 16
Reset Connections           = 12
Current Connections         = 4
Segments Received           = 8617
Segments Sent               = 7641
Segments Retransmitted      = 100

UDP Statistics for IPv4

Datagrams Received = 266
No Ports           = 4
Receive Errors     = 0
Datagrams Sent     = 287
```

C. NMAP

- **Install NMAP**

```
0 upgraded, 7 newly installed, 0 to remove and 1 not upgraded.
Need to get 6,428 kB of archives.
After this operation, 27.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bullseye/main amd64 libblas3 amd64 3.9.0-3+deb11u1 [153 kB]
Get:2 http://deb.debian.org/debian bullseye/main amd64 liblinear4 amd64 2.3.0+dfsg-5 [43.6 kB]
Get:3 http://deb.debian.org/debian bullseye/main amd64 liblua5.3-0 amd64 5.3.3-1.1+deb11u1 [123 kB]
Get:4 http://deb.debian.org/debian bullseye/main amd64 libpcap0.8 amd64 1.10.0-2 [159 kB]
Get:5 http://deb.debian.org/debian bullseye/main amd64 lua-lpeg amd64 1.0.2-1 [33.3 kB]
Get:6 http://deb.debian.org/debian bullseye/main amd64 nmap-common all 7.91+dfsg1+really7.80+dfsg1-2 [4,017 kB]
```

- **Run NMAP**

```
debian@debian:~$ nmap localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-07 10:07 WIB
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
debian@debian:~$ _
```

- **Check Open Ports in repolinux.jti**

```
debian@debian:~$ nmap repolinux.jti.polinema.ac.id
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-07 10:11 WIB
Nmap scan report for repolinux.jti.polinema.ac.id (192.168.60.22)
Host is up (0.00085s latency).
Not shown: 991 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
5357/tcp  open  wsdaapi
8080/tcp  open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
debian@debian:~$ _
```